



AI-DRIVEN PREDICTIVE ANALYTICS FOR BUSINESS GROWTH USING R

Use Case: Real Estate Market Intelligence in Nigeria



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BILIKISU WUNMI OLATUNJI

- **Founder, Lead Data Scientist** - Business Data Laboratory
- **Real Estate Investment Analyst**
- **Founder, Co-organizer** – Abuja R user Group, R-Ladies+ Abuja
- Abuja, Nigeria



THE AFRICAN BUSINESS REALITY

- Many businesses collect data but still make decisions from fragmented signals.
- Real estate decisions involve expensive assets, long cycles, and costly pricing mistakes.
- Nigeria's housing demand and pricing complexity create a strong, relatable use case.
- Predictive analytics can support pricing, investment screening, and market expansion.

From fragmented listings to structured business insight.

DATA → ACTION

**DECISION
SYSTEMS**

LESS GUESSWORK

RISK REDUCTION



BUSINESS PROBLEM

03

Can we use housing market data to estimate property value and support smarter real estate growth decisions?

PRICING QUESTION

What is a defensible market estimate for a property, given its features and location?

INVESTMENT QUESTION

Which listings may be underpriced, overpriced, or unusual enough to deserve closer review?

GROWTH QUESTION

Which markets, segments, or property types look attractive for expansion or prioritization?

Prediction becomes valuable only when it is translated into business decisions.



STAKEHOLDERS

04

Investors

Screen opportunities, compare locations, and speed up due diligence.

Developers

Understand features and locations before allocating capital.

Property tech platforms

Automate price estimation and market insight.

REIT-style analysts

Support property screening and portfolio comparison.

Banks and lenders

Use valuation signals as additional collateral context.

Urban planners

Understand spatial pricing and infrastructure-linked patterns.

One workflow, many decision-makers.



THE DATASET

05

24,326

ROWS

25

STATES

189

TOWNS

7

PROPERTY TITLES

CORE FIELDS

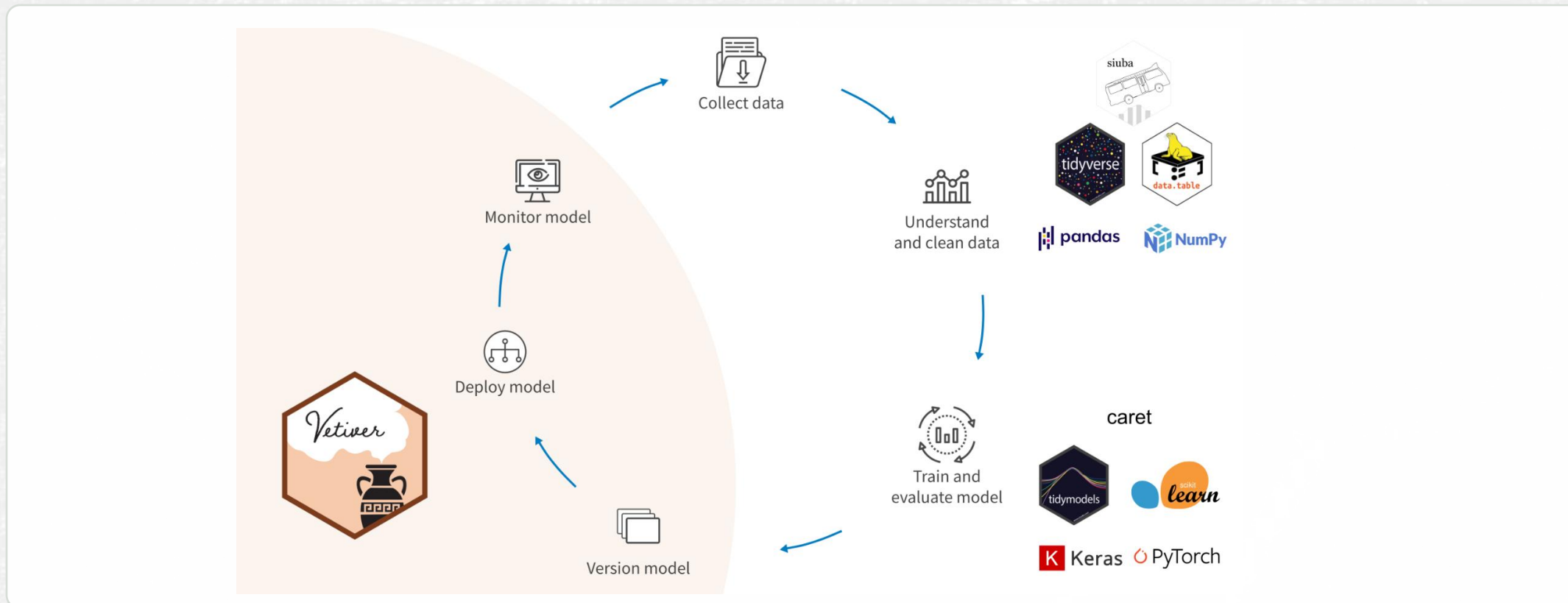
**bedrooms bathrooms toilets parking space
property type town state price**



ML LIFECYCLE

06

The session follows the journey from raw data to decision support.



AI is only useful when the data foundation is credible.

21/21 OK

QUALITY SIGNAL

Validation reduces the risk of training on unreliable inputs.

TRUST

BUSINESS SIGNAL

Credible data improves confidence in pricing and screening outputs.

VALIDATION CHECKS

- 21 *pointblank* checks run before modelling
- Positive price and sensible numeric ranges
- Required fields are present and non-missing
- Categorical values inspected for consistency
- Validation report saved for governance and auditability

Validation report becomes an artifact for stakeholders.



RAW PROPERTY ATTRIBUTES

- Bedrooms
- Bathrooms
- Toilets
- Parking space
- Property type
- Town
- State
- Price



DERIVED BUSINESS SIGNALS

- total_rooms: overall property capacity
- bathroom_bedroom_ratio: comfort signal
- toilet_bedroom_ratio: convenience signal
- parking_per_bedroom: parking adequacy
- location and property encodings: model-ready market signals
- log_price: stable target for skewed house prices

Feature engineering is where raw records become decision-ready variables.

MODEL TRAINING

09

From baseline model to advanced predictive models.

01 Linear regression baseline

A simple,
interpretable
benchmark.

02 Random forest

Captures non-
linear
relationships and
interactions in
tabular data.

03 XGBoost

A boosted tree
model for
stronger
predictive
performance and
ranking.



EXPERIMENT TRACKING

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Logging metrics, parameters, hyperparameters, and artifacts.

WHAT WE LOG

- Experiment names and run context
- Model type and hyperparameters
- Evaluation metrics: RMSE, MAE, R^2
- Predictions and business screening outputs
- Figures, reports, and model artifacts

WHY IT MATTERS

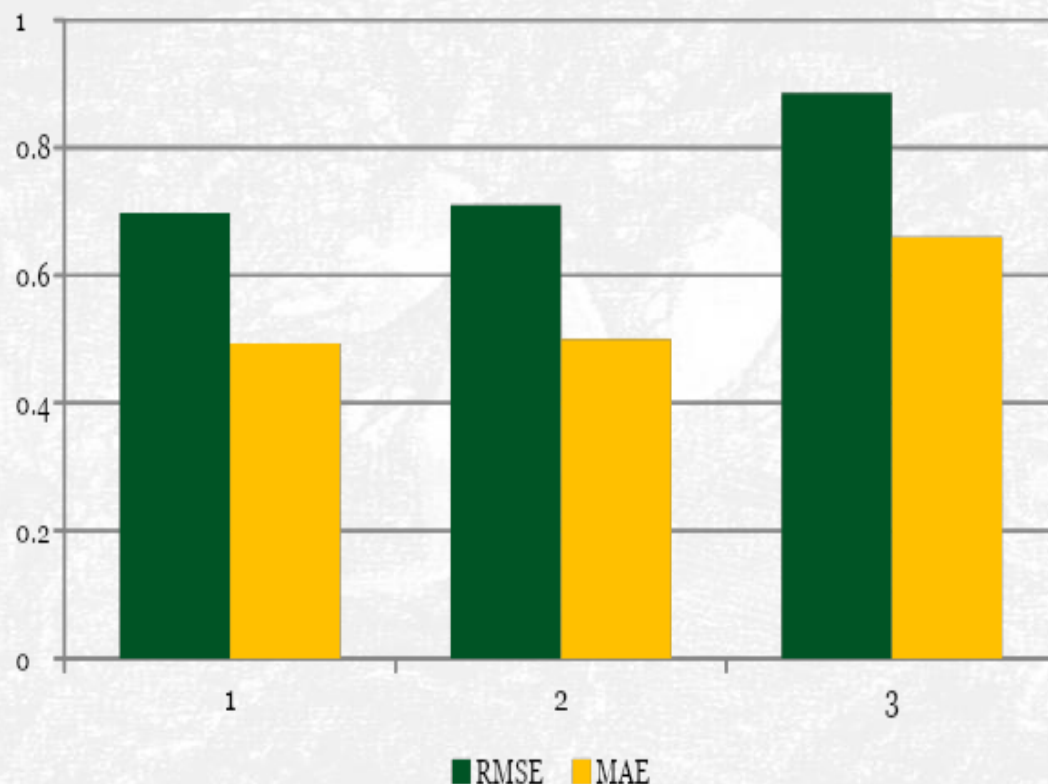
- Improves reproducibility and workshop credibility
- Makes model comparison auditable
- Supports handover to technical teams
- Creates evidence for governance and future improvement



MODEL COMPARISON

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Which model should the business trust?



Best model: XGBoost

RMSE 0.698 | MAE 0.493 | R^2 0.648

- XGBoost was the strongest performer.
- Random forest was a close second.
- Linear regression remains a useful baseline but captured less variation.



BUSINESS INSIGHT LAYER

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Move from prediction to action.

PRICING GAP

Actual price minus predicted price flags unusual listings.

LOCATION RANKING

Summarise market patterns by town and state.

FEATURE IMPORTANCE

Show which variables drive price differences.

MARKET SIGNALS

Separate opportunities from anomalies.

CANDIDATE TABLE INTERPRETATION

- Potential opportunity: below model estimate
- Close to model estimate
- Potentially above model estimate
- Data quality review: unusually below estimate

The model does not say “buy this property now.” It says “this listing behaves differently from the market pattern and deserves investigation.”



WHAT THIS MEANS FOR GROWTH

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Translate technical outputs into business value.

Better pricing

Evidence-based
pricing decisions

Investment screening

Prioritise
candidates for due
diligence

Market prioritisation

Focus on promising
towns and
segments

Reduced decision risk

Spot anomalies and
document
assumptions

Reusable workflow

Repurpose the
lifecycle across
sectors

Predictive analytics creates growth when it helps organisations allocate capital better, reduce risk, and act with more evidence.



THANK YOU

AI-driven predictive analytics is not about replacing business judgment. It is about improving business judgment with evidence.

The power of AI in African markets is in building decision systems that help businesses grow with more evidence, less guesswork, and better accountability.

